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Unit – 4 : Session – 2 : SLO – 2 | Built-in Higher-Order Functions (map, reduce, filter)

■ I. Fill in the Blanks

1. The `filter()` function in Python returns an object that contains elements from the iterable that satisfy a condition (predicate function).
2. A function is pure if it does not have side effects and does not change any external (global) state.

■ II. Multiple Choice

1. `list(map(lambda x: x + 1, [1, 2, 3]))` → `[2, 3, 4]` — map applies the lambda to each element, adding 1 to each.

■ III. Extra Examples

- filter: `list(filter(lambda x: x % 2 == 0, range(10)))` → `[0, 2, 4, 6, 8]`.
- reduce: `functools.reduce(lambda a, b: a * b, [1, 2, 3, 4])` → 24 (factorial of 4).